

A Secure Control Design for Networked Control System with Nonlinear Dynamics under False-Data-Injection Attacks

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Abstract—In a centralized Networked Control System (NCS), all agents share local data with a central processing unit that generates control commands for agents. The use of a communication network between the agents gives NCSs a distinct advantage in efficiency, design cost, and simplicity. However, this benefit comes at the expense of vulnerability to a range of cyber-physical attacks. Recently, novel defense mechanisms to counteract false data injection (FDI) attacks on NCSs have been developed for agents with linear dynamics but have not been thoroughly investigated for NCSs with nonlinear dynamics. This paper proposes an FDI attack mitigation strategy for NCSs composed of agents with nonlinear dynamics under disturbances and measurement noises. The proposed algorithm uses both learning and model-based approaches to estimate agents' states for FDI attack mitigation. A neural network is used to model uncertain dynamics and estimate the effect of FDI attacks. The controller and estimator are designed based on Lyapunov stability analysis. A simulation of robots with Euler-Lagrange dynamics is considered to demonstrate the developed controller's performance to respond to FDI attacks in real-time.

I. INTRODUCTION

Networked control systems (NCS's) have shown promises for their applications in industrial domains, yet, in parallel with the technological advancements boosting their performance, securing them against cyber-physical attacks is also an ongoing research endeavor. During the past several years, researchers focused around securing the NCS's against a major category of cyber-physical attacks, False Data Injection (FDI) attack, where adversary finds vulnerabilities of the system configuration and injects false data into it [1]–[7].

The effect of FDI attacks on NCSs has been studied in the literature extensively [6], [8]–[10]. A class of FDI attacks that can bypass the fault data detection algorithm in Supervisory control and data acquisition (SCADA) systems was introduced by [8]. The work by [9] showed that adversaries could launch FDI attacks against the state estimation system of NCS, with only knowing the system's perturbed model. The work by [10] studied FDI attacks injected on a control system equipped with a Kalman filter. In a similar effort, the most diminutive set of hacker-controlled meters has been identified to conduct an attack that is undetectable [6]. The studies mentioned above confirm that NCSs are vulnerable FDI attacks. Hence, it is crucial to design the next generation of secure controllers in response to FDI attacks.

During the past several years, researchers have observed a progressive development of novel detection algorithms to

detect FDI attacks imposed on an NCS. These techniques can be categorized into learning-based techniques [11]–[17] and model-based techniques [18]–[23]. Model-based techniques require an accurate observer to estimate the system's dynamics. Model-based techniques are suitable for real-time anomaly detection, and they have low computational complexity. However, their absolute dependency on a mathematical model makes them vulnerable to disturbances and model uncertainties. On the other hand, the strength of learning-based techniques and machine learning algorithms to learn and estimate complex nonlinear systems under disturbances make learning-based strategies an invaluable tool for FDI attack detection. Nevertheless, computational complexity makes these techniques inappropriate for real-time detection. Moreover, the stability analysis of learning-based techniques is often more complicated than model-based techniques. Therefore, this paper developed an FDI attack detection and mitigation strategy by combining model and learning-based techniques (for the FDI attack compensation) to increase estimation accuracy while decreasing the computational complexity.

The mitigation of FDI attacks have been investigated in the literature; however, only a few research papers have studied real-time FDI attacks mitigation. The work by [6] utilized state estimation-based observer to mitigate FDI attacks. In another effort, [10] proposed a likelihood ratio test in response to FDI attacks that cannot be identified using threshold-based detection techniques. However, the above-mentioned proposed mitigation techniques rely on an accurate system model. A simultaneous state and input estimation-based algorithm was developed by [24] to mitigate FDI attacks injected into the measurement signals. Nevertheless, the suggested technique considered the FDI attack signal as an unknown input and created a Kalman filter-based state estimator to mitigate the FDI attack. A mitigation algorithm using a Kalman filter for NCS under FDI attacks was proposed in [25]. The algorithm adopted an NN-based architecture to estimate and mitigate FDI attacks in real-time. However, the algorithm is only applicable to systems with linear dynamics.

Despite all the efforts, the mitigation of FDI attacks for nonlinear systems is more challenging. A few approaches have been developed for NCSs with nonlinear dynamics. The work by [26] developed an FDI attack mitigation approach for a class of nonlinear systems using a retrospective cost-based adaptive controller. However, the method is limited to the FDI attacks, which are injected into the actuator signals. A nonlinear-based estimator is designed in [27] to mitigate

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faults injected to state measurements of a control system. The suggested algorithm in [27] applied an NN-based technique to estimate injected faults in real-time. Despite its interesting results, the proposed FDI attack mitigation strategy is not suitable for the second-order nonlinear systems. Furthermore, the stability analysis of the designed controller has not been investigated. In this paper, we designed a Lyapunov-based controller for NCSs with Euler-Lagrange dynamic model, which, to date, has not yet been addressed in the literature.

To summarize, this paper aims at increasing system security by consolidating model-based and learning-based observers for real-time estimation of FDI attacks. The combined utilization of learning-based and model-based observers implements a nonlinear estimator fused with three layers feed-forward neural network that is responsible for real-time detection of the FDI attack. A Lyapunov-based nonlinear controller is designed, which is responsible for securing the NCS under FDI attacks.

The current work builds up and improves the existing knowledge in the security domain of NCS control by framing a design to compensate FDI attacks in real-time. The contribution of the paper is summarized here: 1) the development of a Lyapunov-based control strategy for a class of nonlinear systems under the injection of the FDI attack; 2) the reduction of the computational complexity of learning-based algorithms while increasing the accuracy by incorporating model-based approaches, and 3) the development of stability analysis for the designed control framework.

The rest of the paper is organized as follows: NCS under the FDI attack is formulated in Section II. Section IV illustrates the Lyapunov-based controller design. The developed FDI attack detection algorithm is described in Section V. The stability analysis of the designed controller is provided in Section VI. Finally, Section VII evaluates the performance of the developed nonlinear controller.

II. DYNAMIC MODEL

The structure of a centralized NCS under FDI attacks is shown in Figure 1. The centralized NCS consists of $N \in \mathbb{Z}_{>0}$ agents indicated by $\mathcal{V} \triangleq \{1, 2, \dots, N\}$. The communication topology for the centralized NCS is modeled as an undirected and static graph $\mathcal{G} \triangleq (\mathcal{V}, \mathcal{E})$, in which $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ and $\mathcal{V}$ are the edge set and the node-set respectively. In the centralized NCS, agents transmit their measurable states, x_i , to the control center, where a control signal is generated and is transmitted back to agents. This paper assumed that the graph, $\mathcal{G}$, is connected $\forall t \geq t_0$.

A. NCS under FDI attacks

A class of second-order control affine nonlinear system is considered for the agent. The dynamic model of agent $i \in \mathcal{V}$ is considered as

$$\begin{aligned}\ddot{x}_i(t) &= f_i(x_i, \dot{x}_i, t) + d_i(t) + u_i(t), \\ y_i(t) &= \pi_i(x_i(t), \alpha_{1_i}(t)), \\ \dot{y}_i(t) &= \pi_i(\dot{x}_i(t), \alpha_{2_i}(t)),\end{aligned}\quad (1)$$

where the initial time is denoted by $t_0 \in \mathbb{R}_{\geq 0}$, the number of states for agent i is shown by n_i , $x_i, \dot{x}_i, \ddot{x}_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$

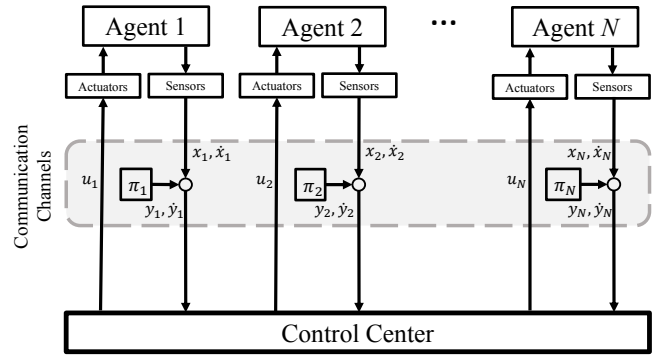


Fig. 1: The illustration of the structure of a centralized NCS under TDS attacks on the network. The vulnerable and secure links are shown with dashed and solid lines, respectively.

denotes the generalized states, $u : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ denotes the generalized input control signal, $f_i(x_i, \dot{x}_i, t) \in \mathbb{R}^{2n_i} \times [0, \infty) \rightarrow \mathbb{R}^{n_i}$ is a known nonlinear C^2 function which is uniformly bounded, $d_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ denotes a bounded exogenous disturbance, $y_i, \dot{y}_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ denote the state measurements under FDI attacks, $\pi_i \in \mathbb{R}^{n_i}$ is a known function, and $\alpha_{1_i}, \alpha_{2_i} \in \mathbb{R}^{n_i}$ represent the FDI attacks.

Assumption 1. The reference trajectory, $x_{d_i} : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$, its first and second derivatives are assumed to be bounded by known positive constants, i.e. $x_{d_i}, \dot{x}_{d_i}, \ddot{x}_{d_i} \in \mathcal{L}_{\infty}$ [28]¹.

Assumption 2. The disturbance is assumed to be continuous and bounded by a known constant, i.e., $\|d_i(t)\| < \bar{d}_i \forall t \geq t_0$, where $\bar{d}_i \in \mathbb{R}_{>0}$ [29].

B. False-Data-Injection Attack

An FDI attack can be generated when an adversary gains access to the communication channels and introduces faults to the data transmitted to the control center from the agents. An adversary's goal is to inject incorrect information to interrupt the system's stable operation while the anomaly detection mechanism remains unaware of the attack. As a result, such FDI attacks can be assumed to be bounded.

In this paper, an FDI attack is modeled by a function π_i , which injects faults into the measured signals as

$$\begin{cases} \pi_i(x_i(t), \alpha_{1_i}(t)) \triangleq x_i(t) + \alpha_{1_i}(t) + \theta_{1_i}(t) \\ \pi_i(\dot{x}_i(t), \alpha_{2_i}(t)) \triangleq \dot{x}_i(t) + \alpha_{2_i}(t) + \theta_{2_i}(t) \end{cases} \quad (2)$$

where $\alpha_{1_i}, \alpha_{2_i} \in \mathbb{R}^{n_i}$ are bounded, unknown, continuous, and time-varying FDI attacks, and $\theta_{1_i}(t)$ and $\theta_{2_i}(t)$ denote Gaussian measurement noises.

Assumption 3. For simplicity of analysis, let $S_{j_i}(t) \triangleq \alpha_{j_i}(t) + \theta_{j_i}(t) \in \mathbb{R}^{n_i}$ for $j = \{1, 2\}$. Therefore, for all $i \in \mathcal{V}$, $S_{1_i}(t)$ and $S_{2_i}(t)$ are bounded and differentiable such that $|S_{j_i}(t)| \leq \bar{S}_{j_i} \forall t \geq t_0, j = \{1, 2\}$, where $\bar{S}_{1_i}$ and $\bar{S}_{2_i}$ are known positive constants.

¹A centralized NCS composed of two-link planar manipulators are considered in this paper.

III. CONTROL OBJECTIVE

This paper's main objective is to design a centralized controller to regulate the agents with the nonlinear dynamics to the desired trajectory despite FDI attacks, additive bounded disturbances, and measurement noises. The error between the desired trajectory, $x_{d_i} : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ and the state, x_i , is defined as

$$e_i(t) \triangleq x_{d_i}(t) - x_i(t), \quad (3)$$

where $e_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ is not measurable during an FDI attack.

The control center only receives the agent's outputs, y_i , and $\dot{y}_i$, which are faulty measurement states during an FDI attack. Therefore, this paper's unique challenge is to ensure accurate tracking, even during an FDI attack. This paper's second objective is to design a nonlinear observer to estimate the NCS's states in real-time. To quantify the accuracy of the observer, the state estimate error, $\tilde{x}_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$, is defined as

$$\tilde{x}_i(t) \triangleq x_i(t) - \hat{x}_i(t), \quad (4)$$

where $\hat{x}_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ denotes the state estimate signals of the agent i .

To facilitate the subsequent stability analysis, two auxiliary tracking errors $r_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ and $\tilde{r} : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ are defined as

$$r_i(t) \triangleq \dot{e}_i(t) + \beta_i e_i(t), \quad (5)$$

and

$$\tilde{r}_i \triangleq \dot{\tilde{x}}_i + \beta_i \tilde{x}_i, \quad (6)$$

where $\beta_i \in \mathbb{R}_{>0}$ is a known user-defined gain.

IV. CONTROL AND OBSERVER DESIGN

A. Control design

The open-loop tracking error for the i^{th} agent can be calculated by taking the derivative of (5) with respect to time, and utilizing (1) to yield

$$\dot{r}_i(t) \triangleq \ddot{x}_{d_i} - f_i(x_i, \dot{x}_i, t) - d_i(t) - u_i(t) + \beta_i \dot{e}_i. \quad (7)$$

The control signal is designed as

$$u_i(t) \triangleq K_i \hat{r}_i(t), \quad (8)$$

where $K_i \in \mathbb{R}_{>0}$ denote a user-defined control gain. Based on the subsequent stability analysis, $\hat{r}_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ is defined as

$$\hat{r}_i(t) \triangleq \dot{\Omega}_i(t) + \beta_i \Omega_i(t) + \hat{S}_i(t), \quad (9)$$

where $\Omega_i \triangleq x_{d_i} - y_i$ is a feedback error signal which is measurable all the time, and $\hat{S}_i \in \mathbb{R}^{n_i}$ is the estimate signal representing the effect of FDI attacks.

Using (1), (2), and (9), $\hat{r}_i$ can be simplified as

$$\hat{r}_i(t) = r_i(t) - \tilde{S}_i(t), \quad (10)$$

where $\tilde{S}_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ denotes the estimation error for the FDI attack and is defined as

$$\tilde{S}_i(t) \triangleq S_i(t) - \hat{S}_i(t), \quad (11)$$

where $S_i(t) \triangleq S_{2_i}(t) + \beta_i S_{1_i}(t)$ denotes the overall effect of FDI attacks on the agent i .

Substituting the control signal (8) into the open-loop tracking error (7), and using (10) results

$$\dot{r}_i = -K_i r_i(t) - e_i(t) + K_i \tilde{S}_i(t) + N_{1_i}(t) + \chi_{1_i}(t), \quad (12)$$

where the auxiliary terms $\chi_{1_i}, N_{1_i} \in \mathbb{R}^{n_i}$ are defined as

$$\begin{aligned} \chi_{1_i}(t) &\triangleq -f_i(x_i, \dot{x}_i, t) + f_i(x_{d_i}, \dot{x}_{d_i}, t) + \beta_i r_i(t) \\ &\quad - \beta_i^2 e_i(t) + e_i(t), \end{aligned} \quad (13)$$

and

$$N_{1_i}(t) \triangleq \ddot{x}_{d_i} - f_i(x_{d_i}, \dot{x}_{d_i}, t) + d_i(t). \quad (14)$$

B. Nonlinear observer Design

The nonlinear observer of agent i is defined based on the subsequent stability analysis as

$$\dot{\hat{x}}_i(t) \triangleq f_i(\hat{x}_i, \dot{\hat{x}}_i, t) + u_i(t) - L_i \Psi_i(t) \quad (15)$$

where $L_i \in \mathbb{R}_{>0}$ denotes known user-defined gains, and $\Psi_i : [t_0, \infty) \rightarrow \mathbb{R}^{n_i}$ is a measurable signal defined as

$$\Psi_i(t) \triangleq \dot{x}_{d_i}(t) - \dot{\hat{x}}_i(t) + \beta_i x_{d_i}(t) - \beta_i \hat{x}_i(t). \quad (16)$$

The closed-loop error for the observer can be calculated by substituting (15) into the time derivative of (6) as

$$\dot{\tilde{r}}_i(t) = -L_i \tilde{r}_i(t) - \tilde{x}_i(t) + \chi_{2_i}(t) + N_{2_i}(t), \quad (17)$$

where the auxiliary terms $\chi_{2_i}, N_{2_i} \in \mathbb{R}^{n_i}$ are defined as

$$\chi_{2_i}(t) \triangleq f_i(x_i, \dot{x}_i, t) - f_i(\hat{x}_i, \dot{\hat{x}}_i, t) + \tilde{x}_i(t), \quad (18)$$

and

$$N_{2_i}(t) \triangleq d_{1_i}(t). \quad (19)$$

By using assumptions 1 and 2 and the Mean Value Theorem (MVT), an upper bound can be identified for the expressions in (13) and (18), i.e. $\|\chi_{1_i}\| \leq \rho_{1_i}(\|z_i\|)\|z_i\|$ and $\|\chi_{2_i}\| \leq \rho_{2_i}(\|z_i\|)\|z_i\|$, where $\rho_{1_i}(\|z_i\|)$ and $\rho_{2_i}(\|z_i\|)$ are globally invertible, positive, and non-decreasing functions [30], and $z_i \in \mathbb{R}^{4n_i}$ is defined as

$$z_i \triangleq [e_i^T, r_i^T, \tilde{x}_i^T, \tilde{r}_i^T]^T. \quad (20)$$

To improve the readability of the subsequent analysis, we omitted the time dependencies in equations.

V. DETECTION OF FDI ATTACK

The developed nonlinear observer (15) includes a neural network-based FDI attack detection algorithm and a state estimator. Due to the fact that the FDI attack function, S_i , is over a non-compact domain, a nonlinear mapping, $M_{s_i} : [t_0, \infty) \rightarrow [0, 1]$, is necessary to map time to a compact spatial domain [31] and is defined as

$$M_{s_i} \triangleq \frac{k_{s_i}(t - t_0)}{k_{s_i}(t - t_0) + 1}, \xi \in [0, 1], t \in [t_0, \infty), \quad (21)$$

where $k_{s_i} \in \mathbb{R}_{>0}$ denotes a user-defined gain. Thus, the FDI attack function, $S_i(t)$, will be mapped into the compact domain, ξ , as described below

$$S_i(t) = S_i(M_{s_i}^{-1}(\xi)) \triangleq S_{M_{s_i}}(\xi). \quad (22)$$

With this nonlinear mapping, $S_{M_{s_i}} : [0, 1] \rightarrow \mathbb{R}^{n_i}$ is defined over a compact domain. Therefore, the Stone-Weierstrass Theorem in [32] is satisfied to estimate the FDI attack function. Now, the FDI attack can be estimated using a neural-network (NN) as

$$S_{M_{s_i}}(\xi) = W_i^T \sigma(V_i^T \delta_i) + \varepsilon_i, \quad (23)$$

where the vectors $W_i \in \mathbb{R}^{(n_n+1) \times n_i}$ and $V_i \in \mathbb{R}^{(n_i+1) \times n_n}$ denote the ideal unknown weights, n_n denotes the neurons number in the hidden layer, $\delta_i \in \mathbb{R}^{(n_i+1) \times 1}$ denotes the inputs, and $\sigma(\cdot) \in \mathbb{R}^{(n_n+1)}$ denotes a vector of activation functions². Furthermore, $\varepsilon_i \in \mathbb{R}^{n_i}$ denotes a bounded signal i.e., $\|\varepsilon_i\| \leq \bar{\varepsilon}_i$, where $\bar{\varepsilon}_i \in \mathbb{R}_{>0}$ is a known positive constant.

The NN estimation of the FDI attack with respect to the spatial domain $[0, 1]$ can be defined by

$$\hat{S}_i \triangleq \hat{W}_i^T \sigma(\hat{V}_i^T \delta_i), \quad (24)$$

where the vectors $V_i \in \mathbb{R}^{(n_i+1) \times n_n}$, $\hat{W}_i \in \mathbb{R}^{(n_n+1) \times n_i}$, are estimates of the ideal weights, and δ_i is defined as

$$\delta_i \triangleq [1, \hat{S}_i^T]^T. \quad (25)$$

Substituting (22), (23), and (24) into (11) results

$$\tilde{S}_i = W_i^T(t) \sigma(V_i^T(t) \delta_i(t)) - \hat{W}_i^T(t) \sigma(\hat{V}_i^T(t) \delta_i(t)) + \varepsilon_i \quad (26)$$

Applying a Taylor's series approximation, the expression in (26) can be written as

$$\tilde{S}_i = \tilde{W}_i^T \sigma(\hat{V}_i^T \delta_i) + \hat{W}_i^T \sigma'(\hat{V}_i^T \delta_i) \tilde{V}_i^T \delta_i + N_{n_i} \quad (27)$$

where

$$N_{n_i} \triangleq \tilde{W}_i^T \sigma'(\hat{V}_i^T \delta_i) \tilde{V}_i^T \delta_i + W_i^T \mathcal{O}(\tilde{V}_i^T \delta_i) + \varepsilon_i, \quad (28)$$

where $\tilde{W}_i \triangleq W_i - \hat{W}_i$ is the outer NN weight error, $\tilde{V}_i = V_i - \hat{V}_i$ is the inner NN weight error, and $\mathcal{O}$ represents higher order terms.

Based on the literature, one can show that N_{n_i} is bounded such that $\|N_{n_i}\| \leq \bar{n}_{n_i}$ for some $\bar{n}_{n_i} \in \mathbb{R}_{>0}$ [34].

Based on the subsequent stability analysis, the updating laws for the NN weights can be obtained as

$$\dot{\hat{W}}_i = \text{proj}(\Gamma_{1_i} K_i \sigma(\hat{V}_i^T \delta_i) (\hat{r}_i + \Phi_i)^T) \quad (29)$$

and

$$\dot{\hat{V}}_i = \text{proj}(\Gamma_{2_i} K_i \delta_i (\hat{r}_i + \Phi_i)^T \hat{W}_i^T \sigma'(\hat{V}_i^T \delta_i)), \quad (30)$$

where $\Gamma_{1_i}, \Gamma_{2_i} \in \mathbb{R}^{n_i \times n_i}$ are positive definite matrices, and the function $\text{proj}(\cdot)$ denotes the projection operator defined in [35]. Furthermore, the signal Φ_i is defined as

$$\Phi_i \triangleq \dot{y}_i + \beta_i y_i - \dot{\hat{x}}_i - \beta_i \hat{x}_i - \hat{S}_i. \quad (31)$$

Remark 1. The $\text{proj}(\cdot)$ function is used to ensure that $\hat{W}_i$ and $\hat{V}_i$ remain bounded for all time.

Remark 2. The effect of FDI attack, S_i , is bounded based on Assumption 3. Furthermore, $\hat{W}_i$ and $\sigma(\hat{V}_i^T \delta_i)$ remain

bounded due to their construction, and therefore, it is obvious from (24) that $\hat{S}_i$ remains bounded. As the result, we can show that the FDI attack estimation error, $\tilde{S}_i$, remains bounded, i.e., $\|\tilde{S}_i\| \leq \bar{S}_i$ for all $t \geq t_0, i \in \mathcal{V}$, where $\bar{S}_i \in \mathbb{R}_{>0}$.

Remark 2, Assumptions 1, and 2 can be used to conclude the following inequalities

$$\|N_{1_i}\| \leq \bar{n}_{1_i}, \quad (32)$$

and

$$\|N_{2_i}\| \leq \bar{n}_{2_i}, \quad (33)$$

where $\bar{n}_{1_i}, \bar{n}_{2_i} \in \mathbb{R}_{>0}$ are known positive constants.

VI. STABILITY ANALYSIS

Let H_i be defined as

$$H_i \triangleq \frac{1}{2} \text{tr}(\tilde{W}_i^T \Gamma_{1_i}^{-1} \tilde{W}_i) + \frac{1}{2} \text{tr}(\tilde{V}_i^T \Gamma_{2_i}^{-1} \tilde{V}_i), \quad (34)$$

where $H_i : [t_0, \infty) \rightarrow \mathbb{R}_{\geq 0}$.

Remark 3. Due to the fact that $\tilde{W}_i$ and $\tilde{V}_i$ are bounded, therefore, H_i is bounded, i.e., $|H_i| \leq H_{i,\max}$, where $H_{i,\max} \in \mathbb{R}_{>0}$ is a known positive constant.

Let $\epsilon_{1_i}, \epsilon_{2_i}, \epsilon_{3_i}, \epsilon_{4_i}, \epsilon_{5_i} \in \mathbb{R}_{>0}$ be user-defined parameters which are chosen to satisfy the following sufficient conditions

$$\begin{aligned} \beta_i &> 0, \\ K_i &> \frac{K_i \epsilon_{1_i}}{2} + \frac{\epsilon_{2_i}}{2} + \frac{\epsilon_{3_i}}{2}, \\ L_{2_i} &> \frac{\epsilon_{4_i}}{2} + \frac{\epsilon_{3_i}}{2} + \frac{K_i \epsilon_{5_i}}{2}, \end{aligned} \quad (35)$$

and let γ_{1_i} and γ_{2_i} be defined as

$$\gamma_{1_i} \triangleq \frac{1}{2} \|z_i\|^2,$$

and

$$\gamma_{2_i} \triangleq \|z_i\|^2.$$

Theorem 1. The controller in (8), FDI attack compensator in (24), and state estimator in (15) ensure globally uniformly ultimately bounded tracking, in the sense that

$$\limsup_{t \rightarrow \infty} \|z_i(t)\| \leq \sqrt{\frac{1}{\gamma_{1_i}} \left(H_{L_{i,\max}} + \frac{\gamma_{2_i} \varphi_i}{\alpha_{3_i}} \right)}, \quad (36)$$

provided that the Assumptions 1-3 and the sufficient conditions in (35) are satisfied.

Proof: Consider $V_i : \mathbb{R}^{4n_i+1} \times [0, \infty) \rightarrow \mathbb{R}_{\geq 0}$, a positive definite, radially unbounded, and continuously differentiable Lyapunov candidate function which is defined as

$$V_i \triangleq \frac{1}{2} e_i^T e_i + \frac{1}{2} r_i^T r_i + \frac{1}{2} \tilde{x}_i^T \tilde{x}_i + \frac{1}{2} \tilde{r}_i^T \tilde{r}_i + H_i. \quad (37)$$

The Lyapunov candidate function satisfies the following inequality

$$\gamma_{1_i} \leq V_i \leq \gamma_{2_i} + H_{i,\max}. \quad (38)$$

²The stacked vector of activation functions are selected to be $\mathcal{O}^2$ [33].

Taking the derivative of (37) with respect to time and substituting in $\dot{e}_i$, $\dot{\tilde{x}}_i$, $\dot{r}_i$, and $\dot{\tilde{r}}_i$ from equations (5), (6), (12), and (17), results

$$\begin{aligned} \dot{V}_i = & e_i^T \left(r_i - \beta_i e_i \right) + r_i^T \left(-K_i r_i - e_i + K_i \tilde{S}_i \right. \\ & \left. + N_{1_i} + \chi_{1_i} \right) + \tilde{x}_i^T \left(\tilde{r}_i - \beta_i \tilde{x}_i \right) + \tilde{r}_i^T \left(-L_i \tilde{r}_i \right. \\ & \left. - \tilde{x}_i + N_{2_i} + \chi_{2_i} \right) + H_i, \end{aligned} \quad (39)$$

Knowing that $\dot{\tilde{V}} = -\dot{V}$ and $\dot{\tilde{W}} = -\dot{W}$, (39) can be written as

$$\begin{aligned} \dot{V}_i = & e_i^T \left(r_i - \beta_i e_i \right) + r_i^T \left(-K_i r_i - e_i + K_i \tilde{S}_i \right. \\ & \left. + N_{1_i} + \chi_{1_i} \right) + \tilde{x}_i^T \left(\tilde{r}_i - \beta_i \tilde{x}_i \right) + \tilde{r}_i^T \left(-L_i \tilde{r}_i \right. \\ & \left. - \tilde{x}_i + N_{2_i} + \chi_{2_i} \right) - \text{tr}(\tilde{W}_i^T \Gamma_{1_i}^{-1} \dot{\tilde{W}}_i) - \text{tr}(\tilde{V}_i^T \Gamma_{2_i}^{-1} \dot{\tilde{V}}_i), \end{aligned} \quad (40)$$

Utilizing (10) and knowing that $\Phi_i = \tilde{r}_i + \tilde{S}_i$ yields

$$r_i^T (K_i \tilde{S}_i) = (\hat{r}_i + \tilde{S}_i)^T (K_i \tilde{S}_i) = (\hat{r}_i + \Phi_i - \tilde{r}_i)^T (K_i \tilde{S}_i) \quad (41)$$

By utilizing (10) and Remark 2, an upper bound for (40) can be obtained as

$$\begin{aligned} \dot{V}_i \leq & -\beta_i \|e_i\|^2 - K_i \|r_i\|^2 + K_i (\hat{r}_i + \Phi_i)^T \tilde{S}_i + K_i \bar{S}_i^2 \\ & + K_i \|\tilde{r}_i\| \|\tilde{S}_i\| + \|r_i\| \|N_{1_i}\| + \|r_i\| \|\chi_{1_i}\| - \beta_i \|\tilde{x}_i\|^2 \\ & - L_i \|\tilde{r}_i\|^2 + \|\tilde{r}_i\| \|N_{2_i}\| + \|\tilde{r}_i\| \|\chi_{2_i}\| \\ & - \text{tr}(\tilde{W}_i^T \Gamma_{1_i}^{-1} \dot{\tilde{W}}_i) - \text{tr}(\tilde{V}_i^T \Gamma_{2_i}^{-1} \dot{\tilde{V}}_i). \end{aligned} \quad (42)$$

Substituting (27) into (42) yields

$$\begin{aligned} \dot{V}_i \leq & -\beta_i \|e_i\|^2 - K_i \|r_i\|^2 + K_i \bar{S}_i^2 + K_i \|r_i\| \|N_{n_i}\| \\ & + K_i (\hat{r}_i + \Phi_i)^T (\tilde{W}_i^T \sigma(\hat{V}_i^T \delta_i) + \hat{W}_i^T \sigma(\hat{V}_i^T \delta_i) \tilde{V}_i^T \delta_i) \\ & + K_i \|\tilde{r}_i\| \|\tilde{S}_i\| + \|r_i\| \|N_{1_i}\| + \|r_i\| \|\chi_{1_i}\| - \beta_i \|\tilde{x}_i\|^2 \\ & - L_i \|\tilde{r}_i\|^2 + \|\tilde{r}_i\| \|N_{2_i}\| + \|\tilde{r}_i\| \|\chi_{2_i}\| \\ & - \text{tr}(\tilde{W}_i^T \Gamma_{1_i}^{-1} \dot{\tilde{W}}_i) - \text{tr}(\tilde{V}_i^T \Gamma_{2_i}^{-1} \dot{\tilde{V}}_i). \end{aligned} \quad (43)$$

Select terms in (43) can be upper bounded by utilizing The Young's Inequality as

$$\|r_i\| \|N_{n_i}\| \leq \frac{\epsilon_{1_i}}{2} \|r_i\|^2 + \frac{\bar{n}_{n_i}^2}{2\epsilon_{1_i}}, \quad (44)$$

$$\|r_i\| \|N_{1_i}\| \leq \frac{\epsilon_{2_i}}{2} \|r_i\|^2 + \frac{\bar{n}_{1_i}^2}{2\epsilon_{2_i}}, \quad (45)$$

$$\|r_i\| \|\chi_{1_i}\| \leq \frac{\epsilon_{3_i}}{2} \|r_i\|^2 + \frac{1}{2\epsilon_{3_i}} \rho_{1_i}^2(\|z_i\|) \|z_i\|^2, \quad (46)$$

$$\|\tilde{r}_i\| \|N_{2_i}\| \leq \frac{\epsilon_{4_i}}{2} \|\tilde{r}_i\|^2 + \frac{\bar{n}_{2_i}^2}{2\epsilon_{4_i}}, \quad (47)$$

$$\|\tilde{r}_i\| \|\chi_{2_i}\| \leq \frac{\epsilon_{5_i}}{2} \|\tilde{r}_i\|^2 + \frac{1}{2\epsilon_{5_i}} \rho_{2_i}^2(\|z_i\|) \|z_i\|^2, \quad (48)$$

$$\|\tilde{r}_i\| \|\tilde{S}_i\| \leq \frac{\epsilon_{5_i}}{2} \|\tilde{r}_i\|^2 + \frac{\bar{S}_i^2}{2\epsilon_{5_i}}, \quad (49)$$

Substituting (29) and (30) into (43) cancels the NN terms. Furthermore, using (44)-(49), (43) can be upper bounded by

$$\begin{aligned} \dot{V}_i \leq & -\beta_i \|e_i\|^2 - \alpha_{1_i} \|r_i\|^2 - \beta_i \|\tilde{x}_i\|^2 - \alpha_{2_i} \|\tilde{r}_i\|^2 \\ & + \frac{1}{2\epsilon_{3_i}} \rho_{1_i}^2(\|z_i\|) \|z_i\|^2 + \varphi_i, \end{aligned} \quad (50)$$

where $\rho_i^2(\|z_i\|) \triangleq \rho_{1_i}^2(\|z_i\|) + \rho_{2_i}^2(\|z_i\|)$, and φ_i is defined as

$$\varphi_i \triangleq \frac{K_i n_{n_i}^2}{2\epsilon_{1_i}} + \frac{\bar{n}_{1_i}^2}{2\epsilon_{2_i}} + \frac{\bar{n}_{2_i}^2}{2\epsilon_{4_i}} + K_i \bar{S}_i^2 + \frac{K_i \bar{S}_i^2}{2\epsilon_{5_i}}. \quad (51)$$

Given on the sufficient conditions defined in (35), the positive constants, α_{1_i} and α_{2_i} can be defined as

$$\alpha_{1_i} \triangleq K_i - \frac{K_i \epsilon_{1_i}}{2} - \frac{\epsilon_{2_i}}{2} - \frac{\epsilon_{3_i}}{2}, \quad (52)$$

$$\alpha_{2_i} \triangleq L_i - \frac{\epsilon_{4_i}}{2} - \frac{\epsilon_{5_i}}{2} - \frac{K_i \epsilon_{5_i}}{2}, \quad (53)$$

The expression in (50) can be upper bounded as

$$\dot{V}_i \leq - \left(\frac{\alpha_{3_i}}{2} - \frac{1}{2\epsilon_{3_i}} \rho_{1_i}^2(\|z_i\|) \right) \|z_i\|^2 - \frac{\alpha_{3_i}}{2} \|z_i\|^2 + \varphi_i, \quad (54)$$

where $\alpha_{3_i} \triangleq \min\{\beta_i, \alpha_{1_i}, \alpha_{2_i}\}$.

Knowing that the Lyapunov function is bounded, the following upper bound for (54) can be obtained

$$\dot{V}_i \leq - \frac{\alpha_{3_i}}{\gamma_{2_i}} V_i + \frac{\alpha_{3_i}}{\gamma_{2_i}} H_{i,\max} + \varphi_i. \quad (55)$$

Solving the above differential equation results

$$\begin{aligned} V_i(t) \leq & V_i(t_0) \exp \left(- \frac{\alpha_{3_i}}{\gamma_{2_i}} (t - t_0) \right) \\ & + \left(H_{i,\max} + \frac{\gamma_{2_i} \varphi_i}{\alpha_{3_i}} \right) \left(1 - \exp \left(- \frac{\alpha_{3_i}}{\gamma_{2_i}} (t - t_0) \right) \right), \end{aligned} \quad (56)$$

$\forall t \in [t_0, \infty)$.

Considering (38), the inequality of (56) can be written in such a way to show the upper bound given in (36). Provided that $z_i \in \mathcal{L}_\infty$, we can conclude that $e_i, r_i, \tilde{x}_i \in \mathcal{L}_\infty$, and equation (8) can be used to show that $u_i \in \mathcal{L}_\infty$.

VII. SIMULATION RESULTS

This section evaluates the developed controller and anomaly detection algorithm's performance for NCSs with nonlinear dynamics under FDI attacks, disturbances, and measurement noise. A centralized NCS composed of two-link planar manipulators is considered for the simulation. For simplicity of the simulation, agents are selected with similar Euler-Lagrange dynamic models [30]. The nonlinear dynamic model of the i^{th} manipulator is described as

$$\begin{aligned} \begin{bmatrix} u_{1_i}(t) \\ u_{2_i}(t) \end{bmatrix} = & \begin{bmatrix} p_{1_i} + 2p_{3_i}c_{2_i} & p_{2_i} + p_{3_i}c_{2_i} \\ p_{2_i} + p_{3_i}c_{2_i} & p_{2_i} \end{bmatrix} \begin{bmatrix} \ddot{x}_{1_i} \\ \ddot{x}_{2_i} \end{bmatrix} \\ & + \begin{bmatrix} -p_{3_i}s_{2_i}\dot{x}_{2_i} & -p_{3_i}s_{2_i}(\dot{x}_{1_i} + \dot{x}_{2_i}) \\ p_{3_i}s_{2_i}\dot{x}_{1_i} & 0 \end{bmatrix} \begin{bmatrix} \dot{x}_{1_i} \\ \dot{x}_{2_i} \end{bmatrix} \\ & + \begin{bmatrix} f_{d1_i} & 0 \\ 0 & f_{d2_i} \end{bmatrix} \begin{bmatrix} \dot{x}_{1_i} \\ \dot{x}_{2_i} \end{bmatrix} + \begin{bmatrix} d_{1_i} \\ d_{2_i} \end{bmatrix}, \forall i \in \{1, 2\}, \end{aligned} \quad (57)$$

where for $i \in \{1, 2\}$, $p_{1_i} = 3.473 \text{ kg} \cdot \text{m}^2$, $p_{2_i} = 0.196 \text{ kg} \cdot \text{m}^2$, $p_{3_i} = 0.242 \text{ kg} \cdot \text{m}^2$, $f_{d1_i} = 5.3 \text{ Nm} \cdot \text{sec}$, $f_{d2_i} = 1.1 \text{ Nm} \cdot \text{sec}$, $c_{2_i} \triangleq \cos(y_{2_i})$, and $s_{2_i} \triangleq \sin(y_{2_i})$. Furthermore, the state measurements under FDI attacks and measurement noises are defined as $y_{j_i} \triangleq \pi_i(x_{j_i}, \alpha_{1_i})$, $\forall j \in \{1, 2\}$, and $\dot{y}_{j_i} \triangleq \pi_i(\dot{x}_{j_i}, \alpha_{2_i})$, $\forall j \in \{1, 2\}$.

The exogenous disturbances are added to both of the agents as

$$\begin{bmatrix} d_{1_i} \\ d_{2_i} \end{bmatrix} = \begin{bmatrix} 0.002\sin(0.1t) \\ 0.001\sin(0.5t) \end{bmatrix}, \forall i \in [1, 2]$$

The desired trajectories for the first agent is selected as

$$x_{d1} = \begin{bmatrix} 50\cos(1.5t)(1 - e^{-0.01t^3}) \text{ deg} \\ 30\cos(1.5t)(1 - e^{-0.01t^3}) \text{ deg} \end{bmatrix},$$

and for the second agent is selected as

$$x_{d2} = \begin{bmatrix} 70\cos(1t)(1 - e^{-0.01t^3}) \text{ deg} \\ 60\cos(1t)(1 - e^{-0.01t^3}) \text{ deg} \end{bmatrix}.$$

The FDI attacks α_{j1} , $\forall j \in \{1, 2\}$ and α_{j2} , $\forall j \in \{1, 2\}$, are applied to the first and second agents and they are modeled as shown in Figures 2 and 3. Table I quantifies the agent's tracking error in the presence of FDI attacks. In the table, the traditional controller refers to a controller without an FDI attack estimator. The position and angular velocity errors are illustrated in Figures 2 and 3. The results clearly show that the developed controller is robust to FDI attacks and additive disturbances. Furthermore, the estimator is able to compensate for the effect of FDI attacks in real-time.

TABLE I: Root Mean Square Error (RMSE) for tracking a desired trajectory while agents are subjected to an FDI attack.

Agent	Proposed controller	Traditional controller
1	0.0405	0.2722
2	0.0532	0.4517

VIII. CONCLUSION

A Lyapunov-based control strategy for NCSs subjected to disturbances, measurement noises, and FDI attacks was developed in this paper. The proposed anomaly detection algorithm was designed based on a combination of model-based and learning-based observers to detect and the effects of FDI attacks in real-time. The Lyapunov stability analysis was utilized to design the update laws for the NN weights. The performance of the controller was evaluated through simulation.

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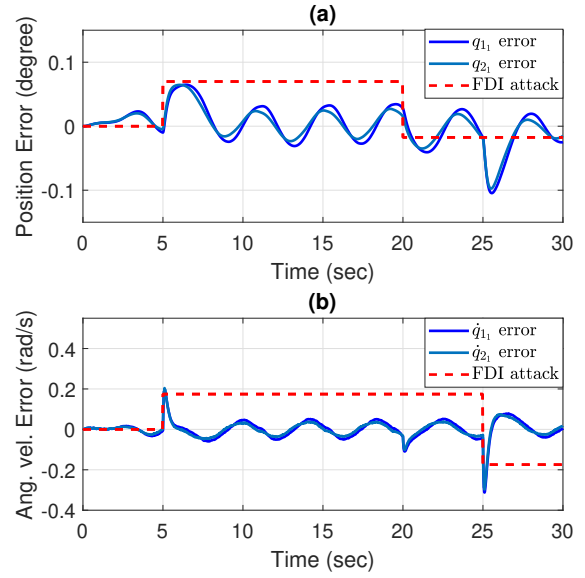


Fig. 2: (a) Position error, (b) Angular Velocity Error for the first agent.

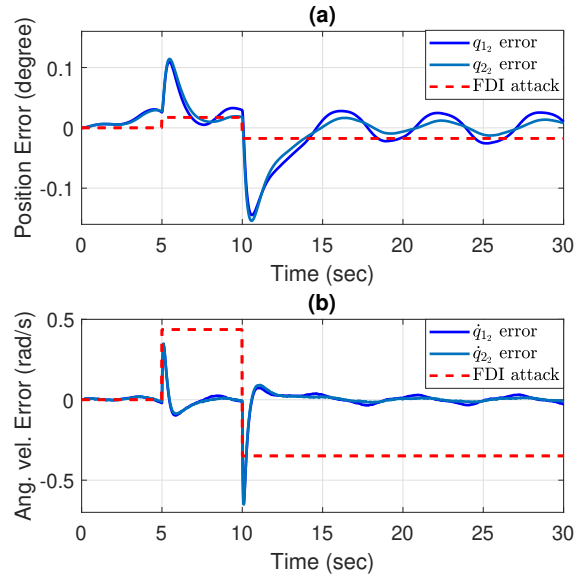


Fig. 3: (a) Position error, (b) Angular Velocity Error for the second agent.

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